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Module 4 Introduction

Lesson 1: Uniform-cost Search & Heuristic Functions

Lesson 2: Greedy Best First Search & Intro to A\* Search

Lesson 3: A\* Search, Continued

Video: A\* Search II

12 mins

Video: Properties of A\* Search

3 mins

Graded Assignment: Learning Check - A\* Search

Submitted

Module Assignment

Your grade: 100%

Next item →

100%

Learning Check - A\* Search

Instructions

Review Learning Objectives

1. True or false: A\* search uses  $h(n)$  to prune the search space so that the use of space is effectively  $O(b^d)$ .  
Ready to review what you've learned before starting the assignment? I'm here to help.

Ⓐ True

Ⓑ False

Ⓒ Correct

Help me practice

Let's chat

The space complexity is  $O(b^d)$ .

Details

1 / 1 point

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2. Consider two consistent and admissible heuristics,  $h_1$  and  $h_2$ , in an A\* search seeking to minimize path costs in a graph.  
Assume ties don't occur in the priority queue. If  $h_1(d) \leq h_2(d)$  for all  $d$ , then...  
To pass, you need at least 80%. We keep your highest score.

Ⓐ A\* search using  $h_1$  will find a lower-cost path than A\* search using  $h_2$ .

Ⓑ A\* search using  $h_2$  will find a lower-cost path than A\* search using  $h_1$ .

Ⓒ A\* search using  $h_1$  will not expand more nodes than A\* search using  $h_2$ .

Ⓓ A\* search using  $h_2$  will not expand more nodes than A\* search using  $h_1$ .

Ⓐ

Ⓑ

Ⓒ

Ⓓ

Correct

Since  $h_2$  dominates  $h_1$ , we know that where efficient,  $h_2$  will not expand more nodes than A\* search using  $h_1$ .

100%

View submission

See feedback

Due Oct 5, 11:59 PM IST

Attempts 3 left (3 attempts every 24 hours)

Try again

3. True or false: A well-designed heuristic would have a value of branching factor close to 1.

Ⓐ True

Ⓑ False

Ⓒ Correct

Smaller branching factor means more nodes will be pruned and make the algorithm more efficient.

1 / 1 point